

3. Project Design Phase

System Architecture

User (browser)



Auth Blueprint (auth.py — /login, /register, /logout, /forgot-password, /reset-password/<token>)



Flask Web Interface (Overview, Smart Recommendation, Suitability Assessment, Crop Library, Research & Policy, Prediction History, About)



Flask Backend (app.py, database.py, utils.py, src/predict.py)



Trained Machine Learning Model (.joblib)



Predicted Crop + Confidence + Report

Modules

Module 1

User Authentication

Module 2

Dataset Collection

Module 3

Data Preprocessing

Module 4

Data Visualization

Module 5

Model Training & Evaluation

Module 6

Model Saving

Module 7

Smart Recommendation (SoilData → Prediction → Report)

Module 8

Suitability Assessment, Crop Library, Research & Policy, Prediction History (search-scoped CSV export)

Input

- Nitrogen (N)
- Phosphorous (P)
- Potassium (K)
- Temperature
- Humidity
- Soil pH
- Rainfall

Output

Recommended Crop with confidence percentage, plus an auto-generated Report (summary and agronomy recommendations) tied to the prediction.

Technologies Used

- Python
- Flask
- Werkzeug (auth/session, password hashing)
- HTML
- CSS
- Bootstrap
- Pandas
- NumPy
- Scikit-learn
- Joblib

Advantages

- Accurate crop recommendation
- Normalized, extensible data model
- Easy deployment
- Simple interface
- Fast execution